

# MATERIAL & PRODUCT SPECIFICATION

|          |                  |       |                                                       |
|----------|------------------|-------|-------------------------------------------------------|
| REF      | QR-2026-0884     | ITEM  | CS-SPR-1140                                           |
| REQ. No. | MR-26-3391 Rev 0 | TITLE | VDSiCr valve-spring wire — helical compression spring |

## PRODUCT DESCRIPTION

Cold-coiled helical compression spring made from oil-hardened-and-tempered chromium-silicon valve-spring wire, for general industrial and valve-actuation duty.

## WIRE MATERIAL SPECIFICATION

EN 10270-2 — VDSiCr (oil hardened and tempered chromium-silicon valve-spring steel wire).  
Related to: ASTM A877 / A877M. Wire diameter: 3.5 mm.

## DELIVERY CONDITION

Oil hardened and tempered; cold coiled; stress-relief tempered after coiling.

## DESIGN & TOLERANCE STANDARDS

Design per EN 13906-1. Cold-coiled compression-spring tolerances per EN 15800 — tolerance grade 2.

## OTHER APPLICABLE DOCUMENTS

WI-COIL-2207; PS-HT-0431; QCP-SPR-118; ESP-0400-12

## CERTIFICATES

Type 3.1 certificates per EN 10204.

## TESTS & SUPPLEMENTARY REQUIREMENTS

- 1 Wire tensile strength 1700–1850 N/mm<sup>2</sup>, per EN 10270-2.
- 2 Decarburization: total  $\leq 1.0\%$  of wire diameter; no complete (ferritic) decarburization. 100% eddy-current surface inspection of wire.
- 3 Surface: maximum defect depth  $\leq 0.5\%$  of wire diameter; free from rust, scale and seams.
- 4 Shot peening: full surface; Almen intensity 0.30–0.45 mmA; coverage  $\geq 100\%$ .
- 5 Set removal (pre-setting / scragging) to solid length; springs delivered preset against further service set.
- 6 Load test (100%): F1 = 320 N  $\pm 7\%$  at L1 = 75 mm; F2 = 640 N  $\pm 7\%$  at L2 = 55 mm, on a calibrated spring tester.
- 7 Geometry: free length L0 = 95 mm; squareness of ground ends  $e \leq 3^\circ$ , per EN 15800 grade 2.
- 8 Surface protection: shot-peened, then zinc-phosphate + dewatering oil. If zinc-electroplated, hydrogen de-embrittlement bake  $\geq 4$  h at 190–220 °C within 4 h of plating.

## INSPECTION / GAUGING

Load-at-length verification on a calibrated spring tester; geometry checked against a reference fixture.

## DIMENSION & QUANTITY

See Purchase Order / drawing 7820.

## NOTES

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